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Variable step size modified P&O MPPT algorithm using GA-based hybrid offline/online PID controller

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ABSTRACT

This paper presents a new modified Perturbation and Observation (P&O) maximum power point tracking algorithm with adaptive duty cycle step using PID controller based on genetic algorithm. The classical P&O MPPT algorithm is widely used in several applications due to its simplicity; however, P&O prone to failure especially when high changes in irradiance, oscillation around the MPP and the convergence speed. To face this challenge and in order to overcome the drawbacks of the classical P&O MPPT, a new method based on variable-step size of modified P&O MPPT method using PID controller tuned by genetic algorithm is presented. The efficiency of the proposed method has been studied successfully using a boost converter connected to a Solarex MSX-60 model. Analysis and comparison with the classical fixed step size P&O and that developed genetic variable step size are presented. The efficiency and improvements of the proposed algorithm in transient, steady-state and dynamic responses, especially under rapidly changing atmospheric conditions, related to ripple, overshoot and response time have been demonstrated. Algorithm robustness was verified using different schemes for temperature and insolation proving its ability to track the maximum power point in case of random and fast changing atmospheric conditions.

Contents

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1. Introduction

The growth of energy demand in the world is steadily increasing and new forms of energy sources must be discovered and developed in order to cover the future demands, in which the world energy demand will increase by 56% from 2010 to 2040 [1]. In this context, many countries have turned to new forms of green energy called "renewable energy" that are currently too expensive and relatively inefficient. Renewable energy is the energy which comes from natural resources such as sunlight, wind, rain, tides and geothermal heat. These resources are renewables and can be naturally replenished. Among the renewable energy sources photovoltaic (PV) energy seems to be the most promising source, due it is available almost everywhere unlike other types such as wind turbines, biomass, geothermal, waves, etc [1,2].

Recently, photovoltaic is growing quickly and becomes increasingly important factor for energetic and economic development, due to these reasons; it achieved a considerable attention as one of the low carbon most promising renewable energy alternatives. The only emissions associated with PV power generation are those from the production of its components. The Photovoltaic's performance depends mainly on the incident irradiation, weather conditions, module temperature, thermal characteristics, module material composition and mounting structure [2,3].

Photovoltaic sources are used today in many applications such as water supply in rural areas, battery charging, mountain cabins, light sources, water pumping meteorological measurement systems, island electrification, satellite power systems, etc. They have the advantage of being maintenance and pollution free but their main drawbacks are high fabrication cost, low energy conversion efficiency and nonlinear characteristics [2–4]. The efficiency of a PV system is affected mainly by three factors:

- The efficiency of the solar panel (9–17%);
- The efficiency of the inverter (95–98%);
- The efficiency of the maximum power point tracking (MPPT) algorithm (which is over 98%)[5,6].

Improvement of MPPT algorithm is the easiest factor due to its cost and it can be used even in equipments which are already in use by updating their control algorithms. Over the last few decades, considerable progress has been made in the MPPT techniques [7–10], these methods can be classified in as online or offline methods and hybrid methods [11,12]. The offline methods does not measure the actual extracted power of the PV panel and they are based on prior knowledge of the photovoltaic panel characteristics and measurements of solar irradiation, such as the short circuit current (I_{sc}) and the open circuit voltage (V_{oc}) [12–14]. These values are employed to generate the control signal necessary for driving the solar cell to its maximum power point (MPP). However, these algorithms cannot detect the maximum power point very accurately, especially during rapid variation in atmospheric conditions [12]. The online methods such as Perturb and Observe (P&O) method [8,11,15], Incremental Conductance (IC) method [8,16–18] and Hill Climbing (HC) [8,19]. The hybrid methods track the MPP with high accuracy, they consisting of two steps: the first step is used for parameter optimization such as PID, fuzzy logic controller and ANN structure. The second step uses the obtained parameter into conventional MPPT methods. Many hybrid methods have been developed such as genetic algorithm-neural networks (GA-ANN) [20,21], optimization of a fuzzy logic controller using particle swarm optimization (PSO-FLC) [22] and genetic algorithm-fuzzy logic controller (GA-FLC) [5,11,23]. Among previous MPPT method, Perturb and Observe, Incremental Conductance and Hill Climbing are the most popular and are widely applied in the MPPT controller due to their simplicity and easy implementation. These methods differ in many aspects such as complexity, sensors required, cost or efficiency, their simplicity, oscillation around the MPP, correct tracking when irradiation and/or temperature change, convergence speed, etc.

Perturb and Observe method based on fixed iteration step size has good performances. However, the main shortcoming of P&O method is that, at the steady state, the operating point oscillates around the MPP resulting in waste some amount of the available energy, the step size is increased for tracking speed-up, the accuracy is decreased and vice versa [24]. To solve the aforementioned drawbacks, modified MPPT algorithms with variable step size have been proposed for some techniques especially the modified Perturb and Observe method, Incremental Conductance MPPT method [18,24,25], in which the step size is automatically adapted according to the inherent PV panel characteristics and improve tracking accuracy as well as tracking dynamics.

A new modified P&O MPPT method with variable step size using PID controller based on genetic algorithm is proposed in this paper. The efficiency of the proposed method has been studied successfully using a boost converter connected to a Solarex MSX-60 model [26]. Results and comparative study between the classical fixed step size P&O and that developed genetic variable step size are presented, three improvements have been demonstrated: response time, overshoot and ripple. In addition, algorithm robustness was verified using different schemes for temperature and insolation proving its ability to track the maximum power point in case of random and fast changing atmospheric conditions.

2. Related works on the use of genetic algorithm in PV MPPT

In the last decade, artificial neural networks (ANNs) [5,8,27–29], fuzzy logic (FL) [8,11,23,30,31], and genetic algorithm (GA) techniques [5,8,20], known as artificial intelligence techniques have been widely used in PV systems process and components, especially under instantaneous climate changes and partially shading conditions. The major reason for this interest is that these techniques are power full and broadly applicable stochastic search and optimization techniques that really work for PV problems that are very difficult to solve by conventional techniques. GAs algorithms, as powerful and broadly applicable stochastic search and optimization techniques, are one of the most widely investigated in the few past years by researcher's community [21,31–33].

From the use of GA concept have resulting fresh research body and applications in PV systems: PV model parameters identification [33–35], PV system sizing [36], PV structure optimization [36] and PV MPPT strategies [31,32,36]. This last application had focused the attention of many researchers and engineers due to its impact on whole system performances. The MPPT, considered as the heart of PV system, adjusts the output power of inverter or DC converter in order to supply reliable energy to the load. The rest of this section constitutes a brief review of the use of GAs in PV system MPPT. Based on the use of genetic algorithm as classification criteria, we can classify this review in two categories named: indirect applications (or hybrid applications) and direct applications.

2.1. Indirect or hybrid applications

In the first category, genetic algorithm techniques are combined with others techniques such as neural networks (NN) or/and fuzzy logic controller (FLC). In this case, the parameters of neural networks and fuzzy logic controller are optimized using genetic algorithm. The GA-ANN and GA-FLC MPPT methods perform better than the conventional P&O method. They exhibit superior transitional characteristics and reduced steady-state fluctuations. In addition, genetic algorithm has been used for the tuning of conventional PID controller parameters.

Akkaya and Kulaksiz have conducted several studies on the use of brushless motor drive for heating, ventilating and air conditioning. In the first study, the brushless motor drive is used as a load for a photovoltaic system [37–39]. The MPPT controller is based on a genetic assisted, multi-layer perceptron neural network (GA-MLP-NN) structure and includes a DC–DC boost converter. Genetic assistance in the neural network is used to optimize the size of the hidden layer. The proposed MPPT controller implemented on DSP, provides an average power increase of 25.35%. In the second study, an ANN was used to determine the reference voltage in real time, dependent upon irradiance and temperature. The dataset used to train the ANN was obtained using experimental measurements, and a relation between the inputs (S and T) and output (V_{MPP}) was established. Due to large dataset used to train the ANN, the GA was used to keep the most decisive data and remove insignificant data [37–39]. In the third one, the application of GA into ANN is regarded as the process of searching for optimal topology for ANN.

In Ref. [40], a genetic algorithm is used to optimize the fuzzy logic controller for the maximum power point tracking under varying atmospheric conditions. By applying the GA, the FLC parameters such as rules base and membership function can be tuned to optimal value where the highest fitness in the last generation will determine the shape of membership function as well as rule base table. Thus, this will improve the performance of the FLC in the optimization process.

In Ref. [41], the authors have presented a new approach based on the genetic algorithm for the tuning of PID parameters in order to optimize the power output of solar panel. The dynamic model is used to design the controller parameters of the conventional PID controller. Due to non-linear characteristics of DC/DC converter, the genetic algorithm is also used to optimize the control parameters of boost converter. Comparison results with conventional techniques show that GA has much faster response than response of the conventional method. In addition, GA designed PID with better rising and settling times performance than the conventional method.

In Ref. [32], Rezaei and Gholamian have proposed an intelligent control technique using fuzzy logic controller optimized by Genetic Algorithm in order to extract the maximum power available at the output of PV module under unstable conditions. Simulation results demonstrated better operation of the optimized fuzzy logic controller under variable weather conditions in comparison with conventional FLC.

In Ref. [42], Seena and Jaimol have proposed a maximum power point tracking using genetic algorithm optimized artificial neural network using PI controller which gives the gating signal to the boost converter is compared with maximum power point tracking using incremental conductance. The performance of MPPT scheme using GA optimized ANN with PI controller outperforms the conventional IC MPPT especially in case of nonlinearities.

2.2. Direct applications

The second category use genetic algorithm directly to estimate parameters required by MPPT method. In addition, this category focused on the development of MPPT controller in case of partial shading considered as one of the most discussed and worked up on concept for the simple reasons that it decreases the power output of the PV array installations and exhibits multiple peaks in the I–V characteristics where conventional algorithms failed to track perfectly the global maximum peak. As a result, the modules have to be reconfigured to get the maximum power output.

In Ref. [43,44] a method, based on the measure of V_{oc} and I_{sc} to compute voltage adjusting the duty cycle is proposed. They showed that proposed control had a strict stability, a good speed and accuracy compared to conventional P&O and IC especially under varying conditions.

Ramaprabha in [45] has introduced a MPPT controller using genetic algorithm for partial shading conditions. The authors present a genetic algorithm based global maximum power point tracking (GMPPT) scheme for a solar photovoltaic array under partial shaded conditions. Due to the presence of multiple peaks in partially shaded PV array, conventional tracking methods fail to work properly, they traps into one of the local peaks. However, in all the cases, GA gives the optimum power (global peak) which is matched with the result of binary search. The comparison results between GA and binary search for different shading scenarios show that the error did not exceed 2%.

Dahmane et al. in [46] have proposed an MPPT strategy that considers the value of short circuit current to generate the current at the maximum power. The current reference is generated by genetic algorithm. The robust control law based on linear matrix inequality (LMI) techniques is then used to track the reference current generated by the GA in order to extract the MPP. The performances of the proposed approach are ensured in simulation, using real profiles of irradiance and temperature measured on platform.

Niranjan et al. in [47] have presented an optimization based approach for total cross tied connected modules in a PV array where the physical locations of the modules remain unchanged while the electrical connections are altered using genetic algorithm as an optimization tool. The resulted connection matrix with the new electrical interconnection fetches the maximum power output using the uniform dispersion of shadow throughout the panel as objective function.

In Shaiek et al. [48], the P&O and the IC techniques, fail to extract the maximum power of the PV panel if the PV generator is partially shaded. So, in these conditions, these techniques fail to extract the global maximum; however, they only detect the first maximum encountered either local or global and regardless of the course. To resolve these problems, a technique based on genetic algorithm is studied and simulated under the same software.

Daraban et al. in [49] have presented a novel MPPT algorithm, based on a modified GA. When photovoltaic systems are affected by partial shading, a global maximum power point tracking algorithm is required to increase the energy harvesting capability of the system. In this algorithm, a P&O algorithm is integrated in the GA function. By embedding a simple P&O MPPT algorithm inside the structure of the GA, the population size and the number of iterations are decreased resulting in a shorter time convergence. The algorithm parameters are optimized and the final solution is provided. The proposed algorithm does not need any preset up configuration and can be directly applied to any type of PV characteristic with an unknown number of local maximum power points. There is no need of a partial shading identification method to avoid losses generated by unnecessary scans.

In Ref. [50], Shankar et al. have proposed a continuous genetic algorithm (CGA) and hybrid particle swarm optimization (HPSO) algorithm are adopted in detecting the global MPP of a partially shaded PV array. From their results, it is observed that the performance of both algorithms, the HPSO and the CGA, is better than either P&O method in finding the global maxima of a partially shaded PV array while P&O alone fails to achieve its objective in case of a partially shaded PV array which may lead to underutilization of the PV array.

In Ref. [51], Liu et al. have presented a literature review including a genetic algorithm as a solution for the tracking of the global maximum power point in case of partial shading. In this

Table 1
Summary of genetic algorithm based maximum power point tracking techniques.

Ref.	Year	Tech.	Remarks
Indirect or hybrid application			
Akkaya et al. [37]	2007	GA & ANN	A genetic assisted, multi-layer perceptron neural network (GA-MLP-NN) structure and includes a DC–DC boost converter. Genetic assistance in the neural network is used to optimize the size of the hidden layer. The proposed MPPT controller provides an average power increase of 25.35%
Larbes et al. [23]	2009	GA & FLC	Genetic algorithms used to find the optimal membership functions. Better performances and robustness under different temperature and irradiance conditions.
Kulaksiz and Akkaya [38]	2011	GA & ANN	GA optimization used to determine neuron numbers in multilayer perceptron neural network. Varying step size decreased response time and oscillations around the MPP. Ideal size for ANN structure with five neurons in hidden layer was obtained.
Ramaprabha et al. [35]	2011	GA & ANN	GA-based, offline-trained ANN provided reference voltage for converter's duty cycle. The error was between 0.05% and 4.46% for different conditions. Improvement suggested by more training of the ANN.
Messai et al. [40]	2011	GA & FLC	Membership functions and rules simultaneously optimized using GA response time in the transition state is shortened, and the fluctuations in the steady state are considerably reduced
Kulaksiz and Akkaya [39]	2012	GA & ANN	GA optimization used to determine neuron numbers in multilayer perceptron neural network. Varying step size decreased response time and oscillations around the MPP. Ideal size for ANN structure with five neurons in hidden layer was obtained.
Mallika et al. [41]	2013	GA & PID	GA-based tuning of PID parameters maximizing the power output of solar panel. In addition, the GA is used to optimize the DC/DC control parameters of boost converter. Comparison results with conventional techniques show that GA has much faster response.
Rezaei and Gholamian [32]	2013	GA & FLC	MPPT Fuzzy logic controller optimized by Genetic Algorithm under unstable conditions. Simulation results demonstrated better operation of the optimized FLC compared to conventional FLC.
Seena and Jomail [42]	2014	GA & ANN-PI	A GA-based optimized ANN with PI controller MPPT is compared to IncCond MPPT. The proposed method outperforms the conventional one especially in case of nonlinearities.
Direct applications			
Hadji et al. [43,44] and	2011	GA	V_{oc} et I_{sc} measured to compute optimal voltage adjusting duty cycle for converter. The proposed controller had a strict stability, good speed, and accuracy compared with PO, and IncCond and also works under varying conditions.
Ramaprabha [45]	2012	GA	GA-based MPPT is used to track the global maximum power point under partial shaded conditions. The GA gives the optimum power in all cases. While conventional techniques fail and traps in one local peak.
Dahmane et al. [46]	2013	GA	MPPT strategy based on short circuit current. The current reference is estimated by genetic algorithm. The robust control law based on LMI techniques is then used to track the reference current generated by the GA. The performances of the proposed approach are ensured in simulation, using real profiles of irradiance and temperature measured on platform.
Shaeik et al. [48]	2013	GA	GA is compared to PO and IncCond techniques in case of multi-peaks. PO and IncCond fail to extract the global maximum power of the PV panel while GA work perfectly.
Daraban et al. [49]	2014	GA	A simple MPPT algorithm (PO) is embedded inside the structure of the GA. The proposed algorithm does not need any preset up configuration and can be directly applied to any type of PV characteristic with an unknown number of local maximum power points.
Shankar et al. [50]	2014	GA	A continuous genetic and hybrid particle swarm optimization algorithms are adopted in detecting the global MPP of a partially shaded PV array. Both methods performs better than either PO method in finding the global MPP.
Deshkar et al. [47]	2015	GA	The paper presents an optimization based approach for Total cross tied (TCT) connected modules in a PV array. The physical locations of the modules remain unchanged while the electrical connections are altered using GA. The new matrix fetches the maximum power output using a uniform dispersion of shadow throughout the panel.
Liu et al. [51]	2015	GA	The paper presents a literature review including GA for tracking the global maximum power point (GMPP) in case of partial shading that causes losses in system output power, hot spot effects, and system safety and reliability problems.

Table 2
Characteristics of MPPT techniques based on genetic algorithm.

Criteria	GA
PV array dependent	No
Convergence Speed	Fast
Periodic tuning	No
Voltage and/or current sensing	Yes
Complexity	Simple
Analog/digital	Digital
Ability to track true maxima	Yes
Initial parameters requirement	Yes
Sensitivity	Moderate

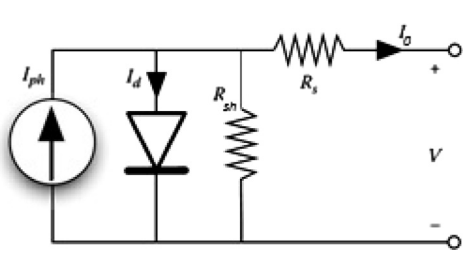


Fig. 1. Equivalent single-diode model of solar cell [2,26].

case, the genetic algorithm working on a population of individuals representing each one a potential solution to the problem, track accurately the global maximum power point while traditional maximum power point tracking techniques fail in local maximum power points.

The principal important points of this brief review of application of genetic algorithm in maximum power point tracking techniques are summarized in Table 1. While Table 2 shows the main characteristics of MPPT techniques based on genetic algorithm [8].

3. PV system modeling

PV cell is an electrical device that converts the energy of light directly into electricity by the photovoltaic effect. A PV module is made of a group of PV cells which are connected together in series and/or parallel circuits to produce higher voltages, currents, and power levels. The obtained energy depends on solar irradiation, the temperature, the spectral characteristics of sunlight, shaded condition, the dirt, and the output voltage of the PV module, etc.

I_o is the PV array output current; I_{ph} is the cell photocurrent that is proportional to solar irradiation; I_d is the current through the diode; R_s and R_{sh} are the series and shunt resistors of the cell respectively.

Table 3
Electrical characteristics of Solarex MSX-60 (1 kW/m², 25 °C) [26].

Description	MSX-60
Maximum power (P_m)	60 W
Voltage Pmax (V_m)	17.1 V
Current at Pmax (I_m)	3.5 A
Short circuit current (I_{sc})	3.8 A
Open circuit voltage (V_{oc})	21.1
Temperature coeff of V_{oc}	$-(80\pm10)$ mV/°C
Temperature co-eff of I_{sc}	(0.065 ± 0.01)% °C
Temperature co-eff of power	(-0.5 ± 0.05)% °C
Nominal operating cell temperature NOCT2	47 ± 2 °C

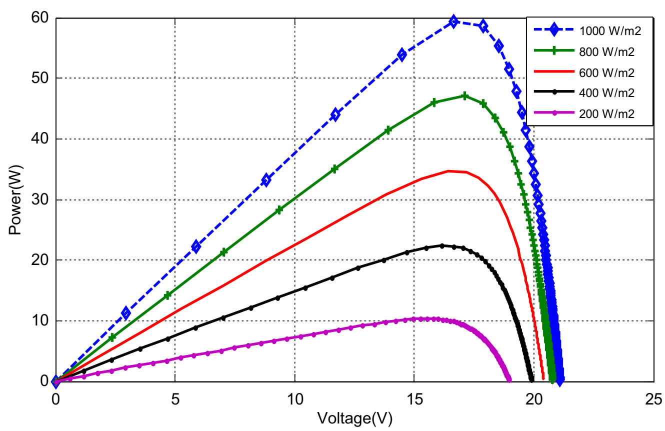


Fig. 2. *P–V* curves for various irradiation levels (Solarex MSX-60, S=0.25, 0.5, 0.75, 1 Sun, T=25 °C).

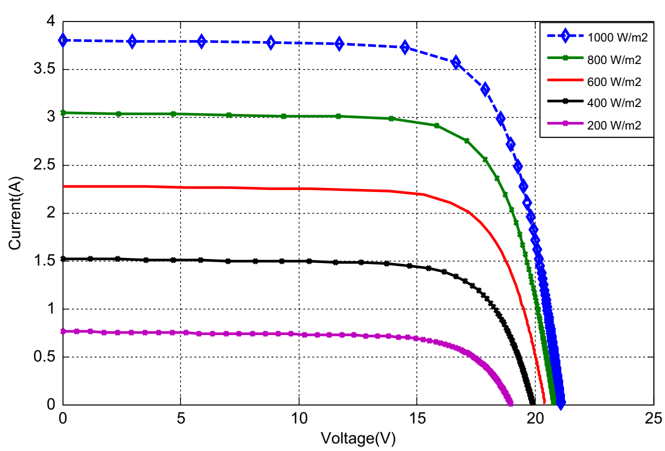


Fig. 3. *I–V* curves for various irradiation levels (Solarex MSX-60, S=0.25, 0.5, 0.75, and 1 Sun, T=25 °C).

Based on single-diode model presented in Fig. 1, the output current of the solar cell can be expressed as [18]

$$I_o = N_p I_{ph} - N_p I_{rs}\left[e^{\left(\frac{q(v+R_S.I_o)}{A.k.T.N_S}\right)} - 1\right] - N_p\left((q(v+R_S I_o))/(N_S.R_{Sh})\right) \quad (1)$$

where

I_{rs} is the cell reverse saturation current;
V is the cell output voltage;
N_s is the number of PV cells connected in series;
N_p is the number of PV cells connected in parallel.
k is the Boltzmann's constant ($1.3806503\times10-23$ J/K);

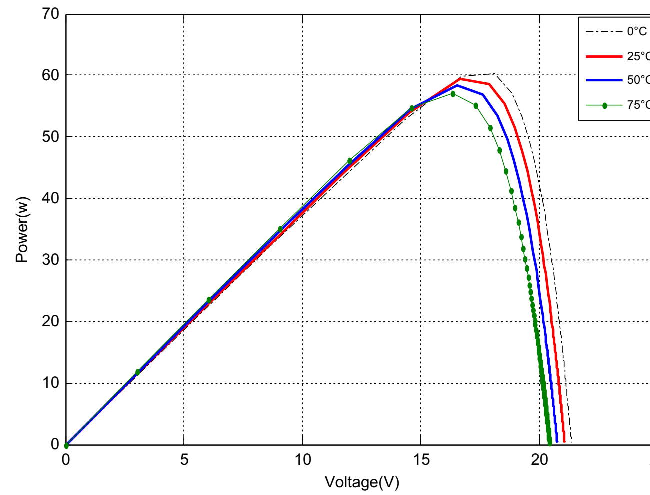


Fig. 4. *P–V* curves for various temperatures. (Solarex MSX-60, S=1 Sun, T=0, 25, 50, and 75 °C).

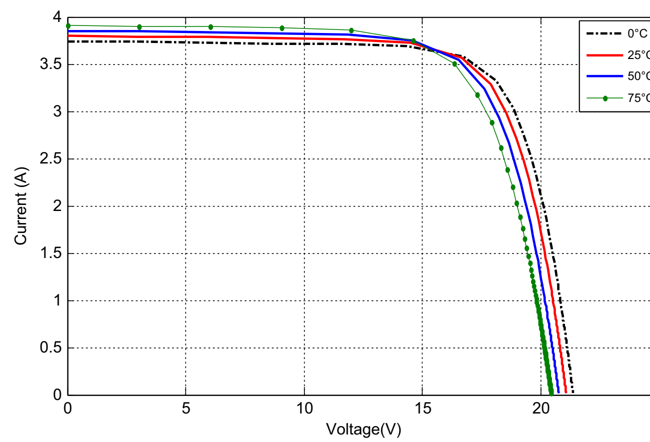


Fig. 5. *I–V* curves for various temperatures. (Solarex MSX-60, S=1 Sun, T=0, 25, 50, and 75 °C).

q is the electron charge ($1.60217646\times10-19$ C);
T is the temperature in Kelvin;
A is the diode ideality constant;

The generated photocurrent I_{ph} is related to the solar irradiation by the following equation:

$$I_{ph} = \left[I_{sc} + k_i(T-T_r)\right](S/1000) \quad (2)$$

where

I_{sc} is the cell short-circuit current at reference temperature;
k_i is the short-circuit current temperature coefficient;
T_r is the cell reference temperature;
S is the solar irradiation in W/m^2.

The cell's saturation current is varies with temperature according to the following equation:

$$I_{rs} = I_{rr}[T/T_r]^3\exp\left((q.E_G)/(k.A)\left[(1/T_r)-(1/T)\right]\right) \quad (3)$$

where

I_{rr} is the reverse saturation at T_r;
T_r is the cell reference temperature;
E_G is the band-gap energy of the semiconductor used in the cell.

For simulations and analysis of proposed method, the Solarex MSX-60 photovoltaic module with 36 solar cells was chosen [26]. The electrical characteristics of the Solarex MSX60 module for $S=1000$ W/m2 and $T=25$ °C are given in Table 3. The resultant I–V and P–V curves are shown in Figs. 2–4 and 5.

Figs. 2 and 3 show the P–V characteristics and I–V voltage characteristics of the PV cell for different irradiation levels (1000, 800, 600, 400, and 200 W/m^2) at constant temperature ($T=25$ °C), respectively.

Figs. 4 and 5 show the effect of temperature variation ($T=25$, 50, 75, and 100 C°) at constant irradiance ($S=1000$ W/m^2) on P–V and I–V characteristics respectively.

4. Modified P&O MPPT

Recently, many MPPT techniques have been developed and improved continuously, Perturb and Observe method is widely applied in the MPPT controllers due to its simplicity and easy implementation. The P&O MPPT algorithm is based on the disturbance of the output voltage $V(k)$ of the PV panel and the corresponding output power $P(k)$, which is compared with the previous perturbing cycle $P(k-1)$. If the power increases, keep the next voltage change in the same direction as the previous change. Otherwise, change the voltage in the opposite direction as the previous one, when the stable condition is reached the algorithm oscillates around the peak power point. Fig. 6 shows the flowchart of the P&O method.

This method has several drawbacks such as slow tracking speed and oscillations about maximum power point, making it less favorable for rapidly changing environmental conditions. Using a larger perturbation steps, the maximum power point is reached quickly, but, the power loss caused by perturbation in the steady state will also increase. With a smaller perturbation step can reduce the power loss caused by perturbation in the steady state but will slow down the tracking speed.

5. Proposed GA-based variable step P&O MPPT

The power provided by the PV array varies with solar irradiance and temperature, since these parameters influence the I–V characteristics of solar cells. In order to optimize the energy transfer from the PV array to the load, it is necessary to force the working point to be at the maximum power point. In this way, the MPPT controllers drive the converter through the changing of the duty cycle of the DC/DC converter. In most of the cases, the controller produces a reference voltage for a Pulse Width Modulation (PWM) generator to produce the appropriate pulses. Most control schemes use the P&O fixed step technique because it is easy to implement but the oscillation problem is unavoidable. To overcome this drawback, a new genetic algorithm based variable step P&O algorithm is proposed. The P&O variable step is optimized driving PID controller tuned by genetic algorithm. The GA select the K_p, K_i and K_d PID parameters value that optimizes the power transfer though the DC/DC converter. The system structure block is shown in Fig. 7.

5.1. Genetic algorithm

In biological evolution, species are positively or negatively selected depending on their relative success in surviving and reproducing in their current environment. Differential survival and variety generation during reproduction provide the engine for evolution. These concepts have metaphorically inspired a family of algorithms known as evolutionary computation. The genetic algorithms are a family of computational models inspired by evolution. They are parallel global probabilistic search techniques based on the principle of population genetics [52,53]. These algorithms encode a potential solution to a specific problem on a single chromosome and apply recombination operators to them so as to preserve critical information. GAs are often viewed as function optimizers, although the range of problems to which GAs have been applied is quite broad. The major reason for GAs popularity in

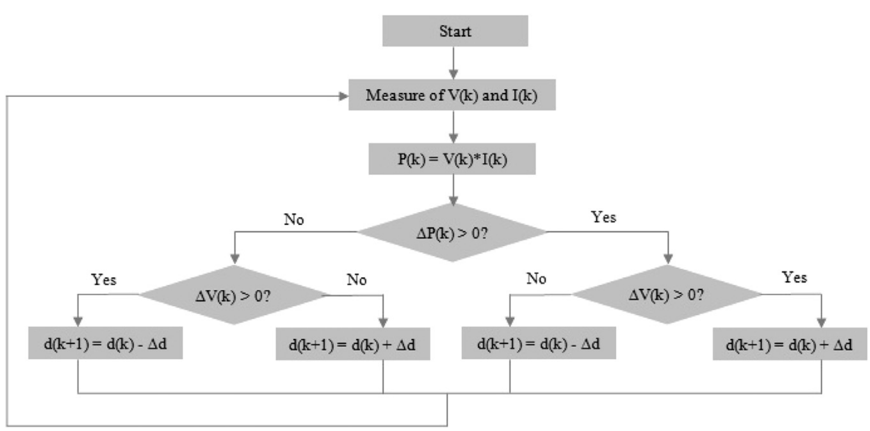


Fig. 6. Flowchart of the P&O method.

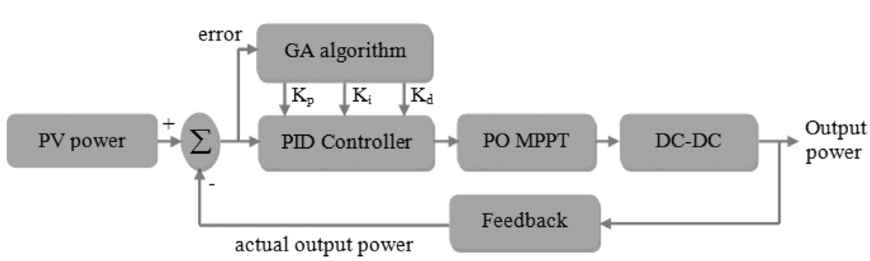


Fig. 7. The block diagram of proposed Genetic Variable Step P&O MPPT.

various search and optimization problems is its global perspective, wide spread applicability and inherent parallelism. GA starts with a number of solutions known as chromosomes. These solutions are represented using a string coding of fixed length. After evaluating each chromosome using a fitness function and assigning a fitness value, three different genetic operators (selection, crossover and mutation) are applied to update the population. The selection is applied on a population and forms a mating pool. The crossover operator is applied next to the strings of mating pool, it picks two strings from the pool at random and exchanges some portion of the strings between them. The mutation operator changes 1–0 and vice versa. An iteration of these three operators is known as a generation. If a stop criterion is not satisfied this process repeats. This stop criterion can be defined as reaching a predefined time limit or number of generations or population convergence. The overview of genetic algorithm working principles of a simple GA is shown in Fig. 8 [53,54,55].

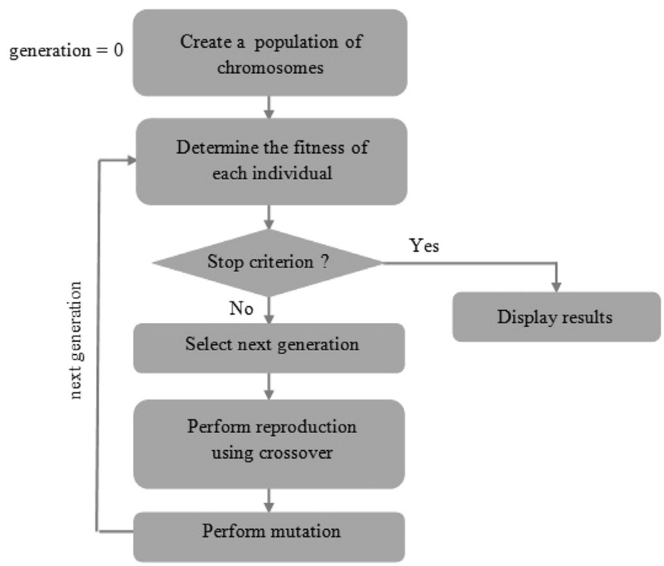
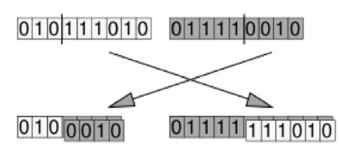


Fig. 8. The block diagram of basic genetic algorithm.

5.2. PID controller genetic algorithm tuning

The PID controller genetic algorithm tuning procedure flowchart is shown in Fig. 9.

The PID parameters K_p, K_i and K_d are encoded into binary strings known as chromosomes. The length of strings is set to 48 ($16+16+16$ for the three controller gains). The selection is applied on a population of chromosomes and forms a mating pool. The purpose of parent selection in a GA is to give more reproductive changes to those individuals that are the fit. In our case we use the roulette wheel parent selection. The crossover operator is applied next to produce new chromosomes. Like in nature, crossover produces new individuals having some parts of both parent's genetic material. For this matter, we apply a single point crossover.



Mutation is a genetic operator used to maintain genetic diversity from one generation of a population of genetic algorithm

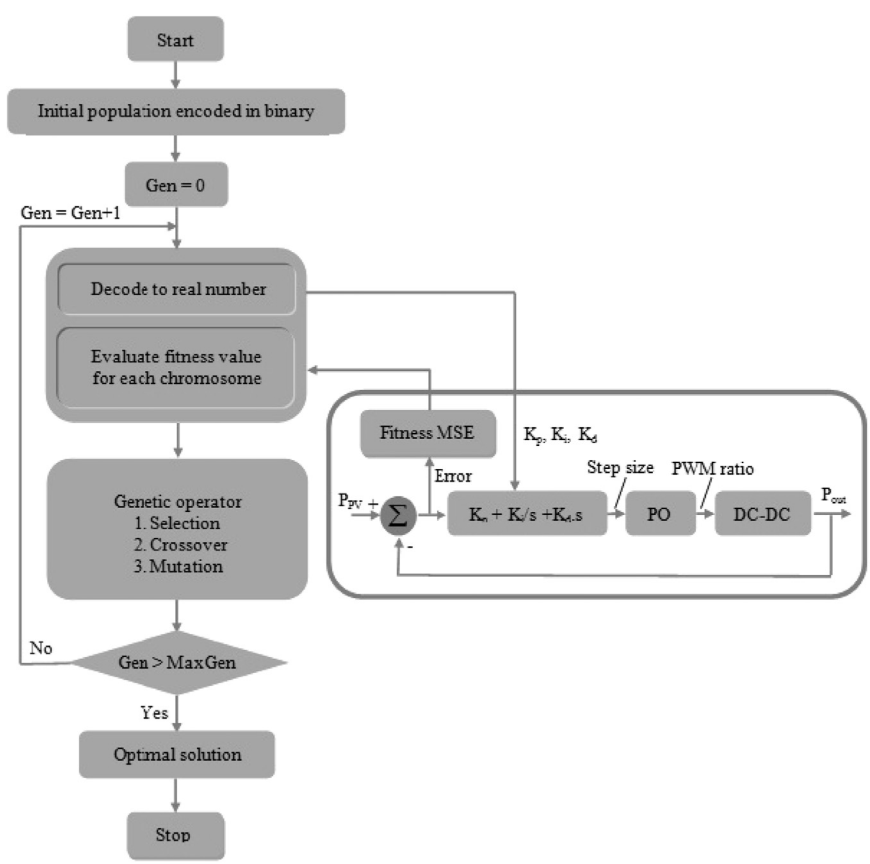


Fig. 9. Detailed block diagram of the proposed algorithm.

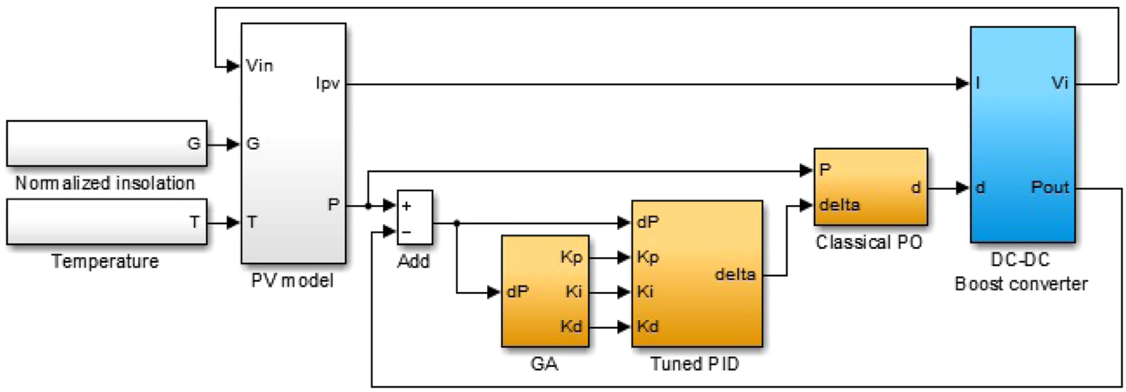


Fig. 10. Matlab/Simulink Variable step PO-GA MPPT implementation.

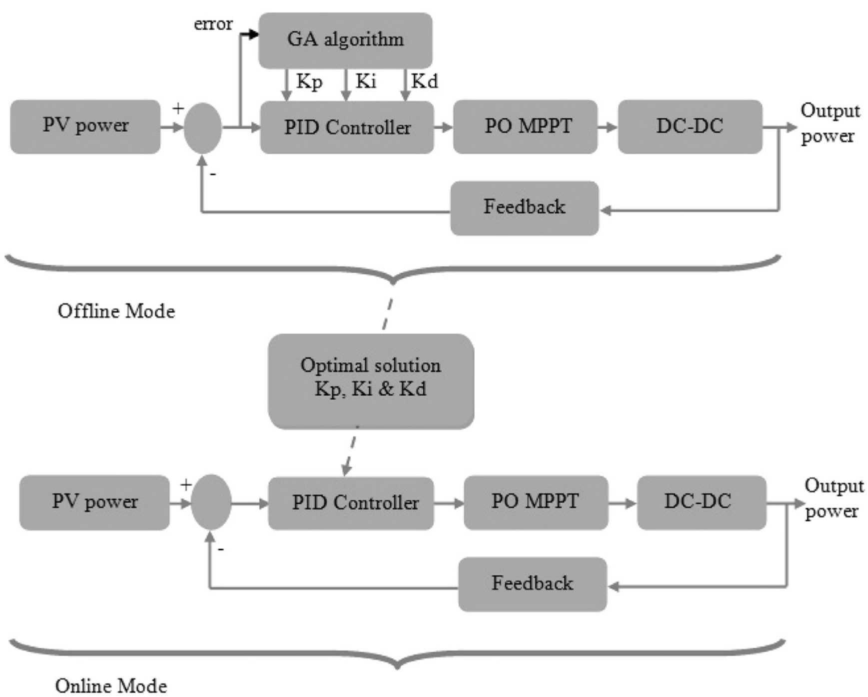


Fig. 11. Proposed system operating modes: offline mode (top) & online mode (bottom).

Table 4
GA parameters.

Description	Parameters
Population size	20
Maximum iteration	50
Crossover probability	0.5
Mutation probability	0.01
Number of bits per chromosome	16
$\alpha=\beta$	0.5

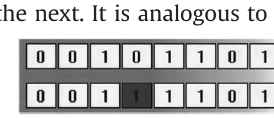
Table 5
Test pattern signals.

Irradiation (W/m^2)	Time (s)
1000	0–2.5
800	2.5–5
600	5–7.5
400	7.5–10

Table 6
Optimum set of PID controllers gains

PID optimum set	K_p	K_i	K_d
PID Set 1	1.173500	5.103530	0.327901
PID Set 2	1.616800	6.979500	0.952450
PID Set 3	1.573600	8.123800	0.320580
PID Set 4	0.554060	8.880600	0.174930
PID Set 5	0.616220	6.743300	0.265460

chromosomes to the next. It is analogous to biological mutation.



The fitness of each chromosome is evaluated by converting its binary string into a real value which represents PID gains. Each set of PID parameters is passed to PID controller in order to compute a complete response of the system as described in Fig. 9 and its initial fitness value is computed using the below defined objective

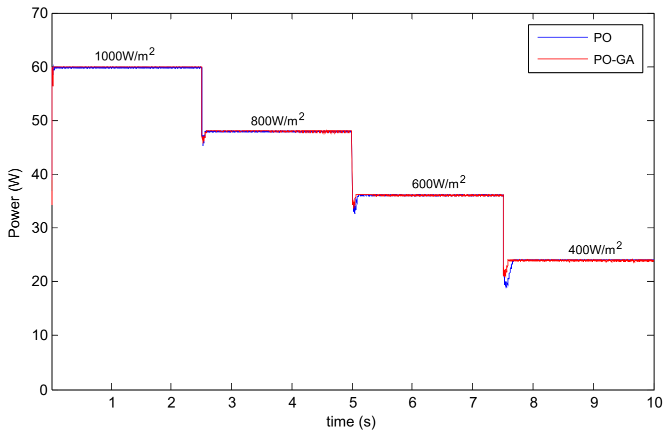


Fig. 12. Output power. Fixed step PO (blue curve), GA-based variable step PO-GA (red curve). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

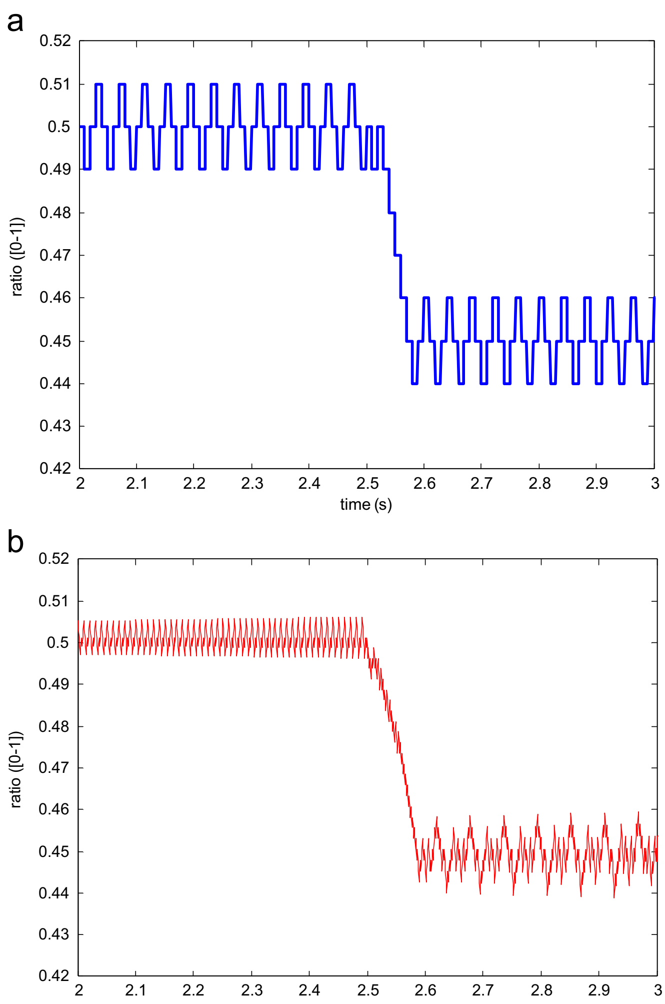


Fig. 13. DC/DC PWM ratio: (a) fixed step PO, and (b) GA-based variable step PO-GA.

function [56]:

$$F = \alpha.\text{overshoot} + \beta.\text{ISE} \tag{4}$$

where

$$\text{ISE} = \int_0^{\tau} (P_{ref} - P_{out})^2 dt) \text{ and overshoot} = \max(P_{out}) - P_{ref}$$

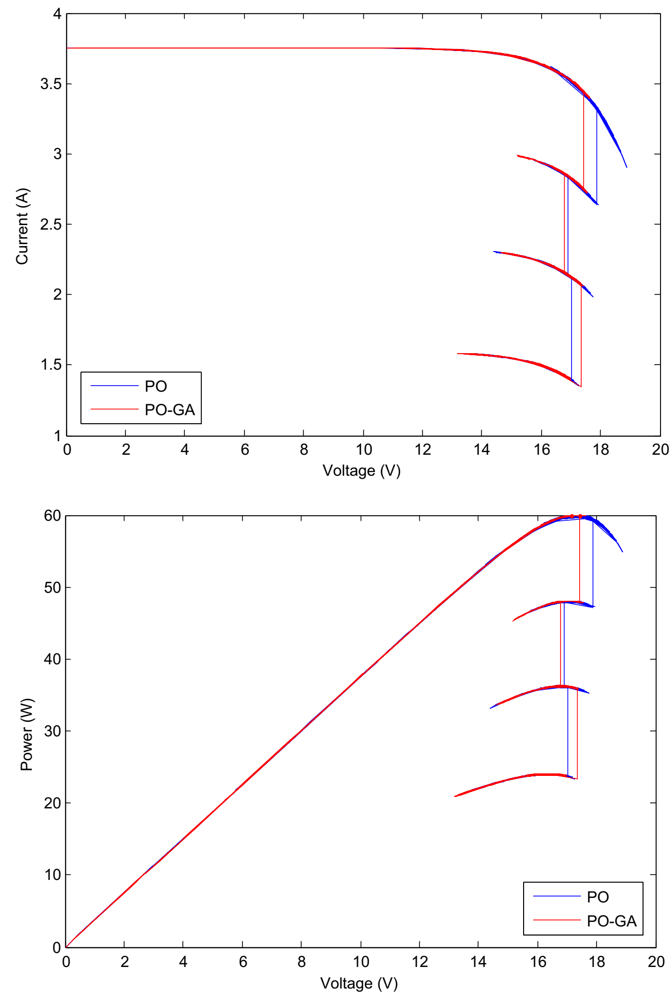


Fig. 14. MPP Tracking. Fixed step PO (blue curve), GA-based variable step PO-GA (red curve). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

In our case, we have no preference between the two objectives, we choose ($\alpha = \beta = 0.5$).

The process will go through GA steps sequentially and repeat until the end of generations reached where the best fitness is achieved and optimal PID parameters found.

5.3. PO-GA MPPT implementation

Designing an optimum PID controller ensures improved variable step size for P&O MPPT by minimizing the performance index. To illustrate the importance of proposed PO-GA algorithm, the model of the whole system described by Fig. 10 designed using Matlab/Simulink software is considered. It is composed of PV panel operating at variable temperature and irradiation, PID controller tuned by GA to compute the variable step size needed by P&O MPPT controller that drives the duty cycle of DC/DC converter.

The system operates in two modes: (1) The offline mode required for testing different set of PID parameters to find the optimal controller, (2) The online mode that uses the found optimal PID to track the MPP point. Fig. 11 shows the two modes [54].

The simulated GA algorithm parameters used to initialize the GA algorithm parameters and generating an initial random population of individuals representing the PID parameters (K_p, K_i and K_d) are defined in Table 4. The GA is implemented with small

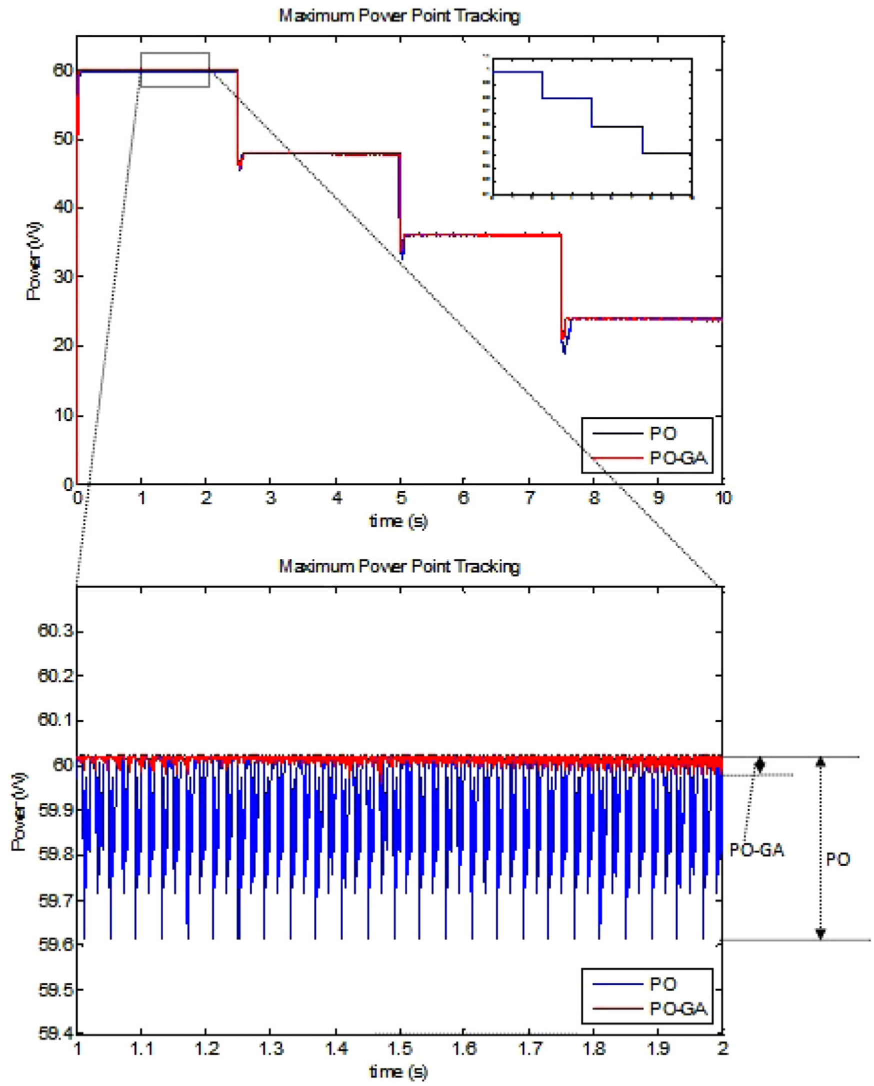


Fig. 15. Maximum Power Point. Fixed step PO (blue curve), GA-based variable step PO-GA (red curve) (Ripple improvement). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

population size. This requirement is important in practice in order to allow the controller to be optimized as fast as possible. In this study, the size of initial populations is set to be 20 and the number maximum of generations $G=50$.

6. Results and discussions

Having in mind that irradiation can change quickly due to weather conditions or passing clouds and in order to evaluate the performance of the proposed algorithm compared to P&O classical algorithm, four irradiation steps signals were simulated using MATLAB/Simulink software as described by Table 5.

6.1. Offline mode tests

The GA is a mathematical optimization algorithm with global search potential. The GA adopts a populational unit of analysis, wherein each member of the population encodes a potential solution to PID controller gains tuning problem. By the way, in offline mode, after all realized simulations, the GA provides a list of optimum sets of PID controller gains. The optimum PID sets of Table 6 give the same global objective with different performances (ripple, overshoot and response time).

6.2. Online mode tests

The online mode testing has been made with the PID set 1, which gives the best results in terms of transient, steady-state and dynamic responses of the algorithm, especially under rapidly changing atmospheric conditions. Fig. 12 shows maximum power point tracking at the output of the DC/DC boost converter corresponding to the input irradiation scheme defined in Table 4 (graph in left bottom corner of Fig. 12). We can see that in both cases (P&O or PO-GA algorithms) the output power follows the insolation variation at the input of the PV panel.

The Fig. 13 below shows the duty cycle provided by both conventional P&O (fixed step size) and proposed PO-GA algorithms (variable step size).

6.2.1. MPPT tracking

The MPPT point tracking by both algorithms (P&O and PO-GA) is shown by Fig. 14 (top: IV characteristics and bottom: PV characteristics).

We can clearly see that the race of the MPPT point in most cases is less important for the proposed PO-GA algorithm compared to the conventional P&O algorithm.This is due to the

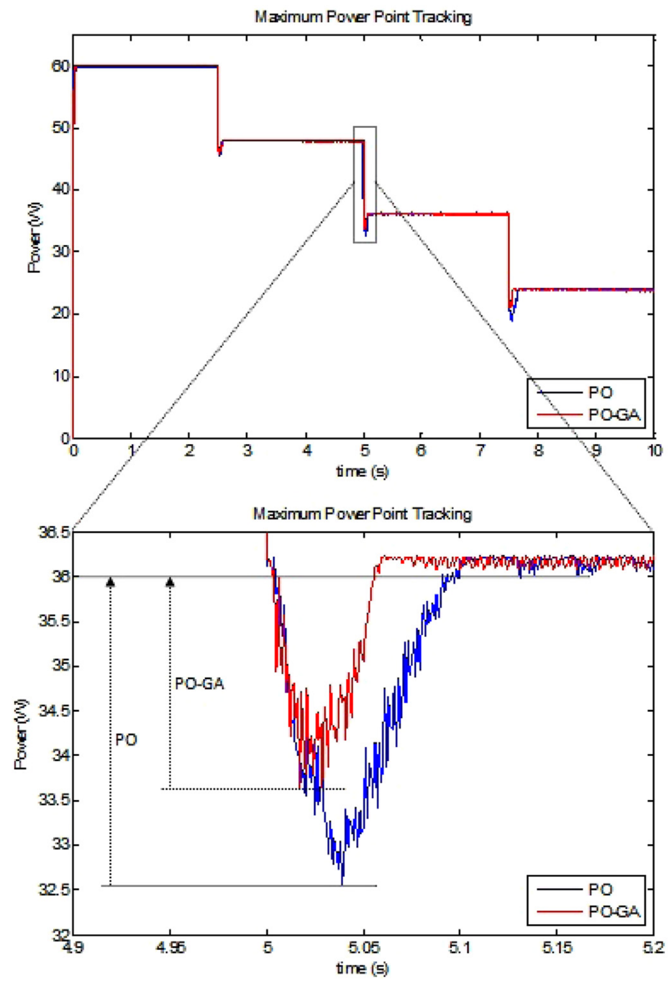


Fig. 16. Maximum Power Point Tracking. Fixed step PO (blue curve), GA-based variable step PO-GA (red curve) (Overshoot improvement). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

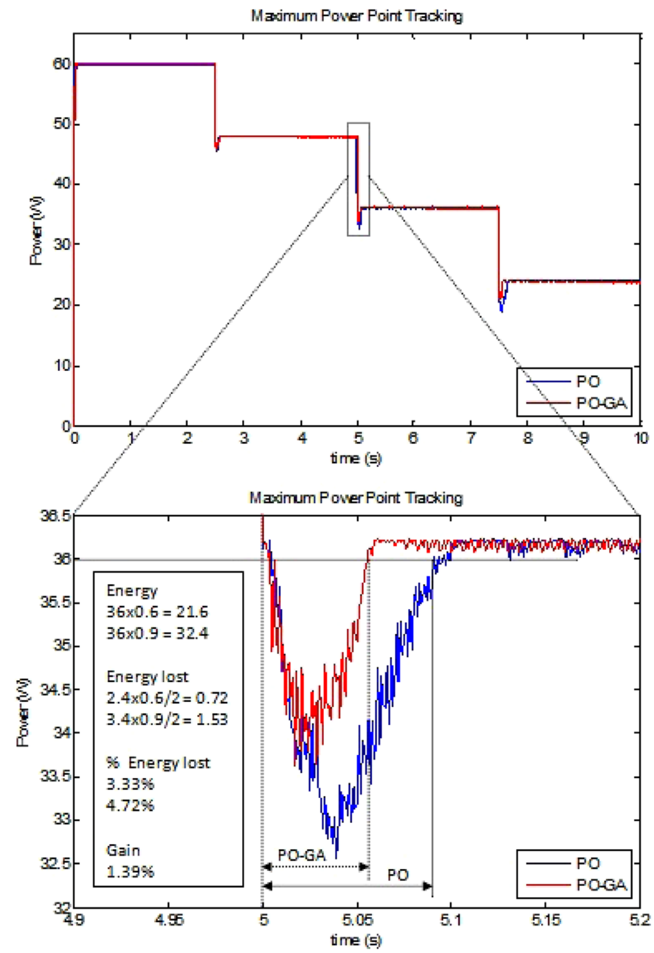


Fig. 17. Maximum Power Point Tracking: fixed step PO (blue curve), GA-based variable step PO-GA (red curve) (Response time improvement). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

instability and the oscillation of the P&O algorithm especially around the MPPT point.

6.2.2. Ripple

The improvement of PO-GA algorithm regarding ripple is undeniably clear in Fig. 15 (bottom).

The ratio between the signals issued by P&O and PO-GA algorithms, respectively is more than 13 times (0.67% for PO instead of 0.05% for PO-GA).

6.2.3. Overshoot

The overshoot due to irradiation changing from 800 W/m^2 to 600 W/m^2 is less important in case of PO-GA algorithm (1.3%) than in case of P&O conventional algorithm (9.69%). This improvement is considered using only the pic value that is the maximum, knowing that the duration of this peak is less in the case of PO-GA proposed algorithm compared to the conventional P&O algorithm (Fig. 16).

6.2.4. Response time

The maximum power point tracking comparison is shown in Fig. 17. The proposed algorithm shows a significant improvement in the time interval corresponding to the period of rapidly changing atmospheric conditions. In addition, the proposed algorithm shows a better transient response which can be seen in the time interval 5 s $< t <$ 5.06 s.

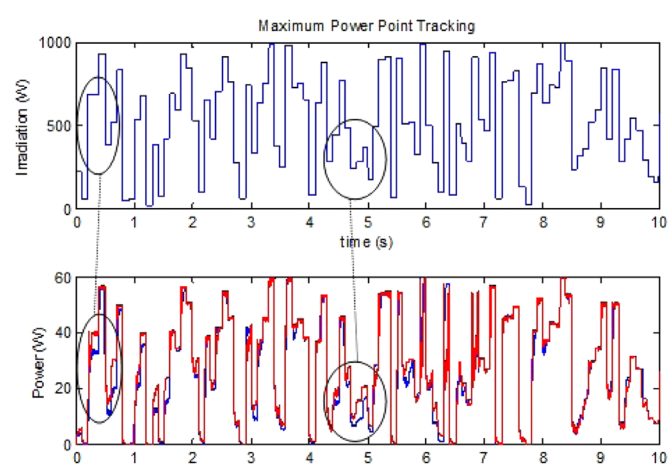


Fig. 18. Maximum Power Point Tracking in case of random S input and fixed temperature (T=25 °C). Fixed step PO (blue curve), variable step PO-GA (red curve). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The response time for the PO-GA is equal to 60 ms while it is equal to 90 ms for the P&O conventional algorithm. The energy

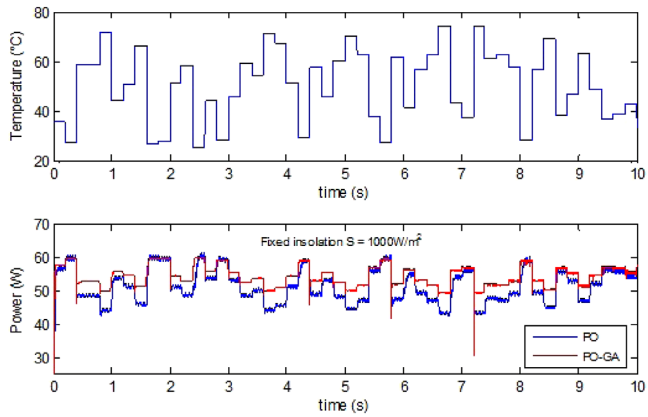


Fig. 19. Maximum Power Point Tracking in case of random temperature T input and fixed insolation ($S=1000\ \mathrm{W/m^2}$). Fixed step PO (blue curve), variable step PO-GA (red curve). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

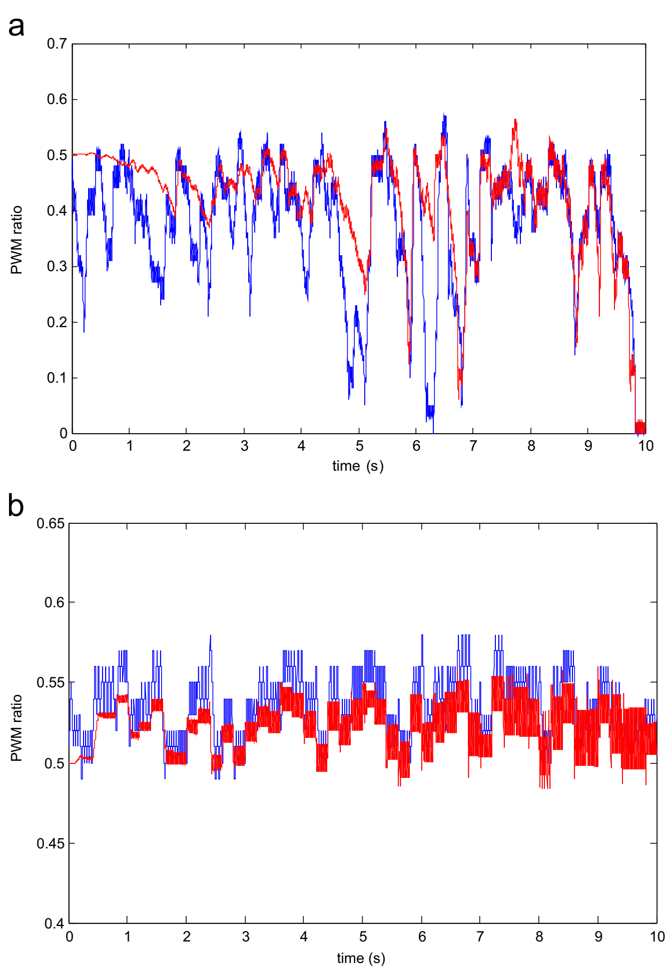


Fig. 20. DC/DC PWM ratio. (a) Random insolation S with fixed temperature ($T=25\ °C$), and (b) random temperature with fixed insolation ($S=1000\ \mathrm{W/m^2}$). Fixed step PO (blue curve), variable step PO-GA (red curve). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

lost is 3.33% and 4.72% for proposed PO-GA and conventional P&O algorithms, respectively. From the point of view gained power, by applying the proposed PO-GA algorithm, we have approximately recovered 1.4% of the lost energy. Noted that the 1.4% energy recovered, in this case, is considered only for one changing event (from 800 $\mathrm{W/m^2}$ to 600 $\mathrm{W/m^2}$). This gives a good idea about the realized energy recovered considering the changing nature of environment conditions and insolation.

Table 7
Improvements using the PO-GA proposed algorithm

Parameter	Conventional PO	Proposed PO-GA	PO-GA improvement (%)
Ripple (%)	0.67	0.05	**92.54**
Overshoot (%)	9.60	1.30	**86.58**
Resp. time (ms)	90.00	60.00	**33.33**
Energy lost (%)	4.72	3.33	**29.45**

6.3. *Robustness and reliability tests of the proposed PO-GA algorithm*

To prove the robustness and reliability of our proposed algorithm, we test its ability to track the MPP in different conditions varying insolation and temperature. In the first test, we use a random insolation scheme with fixed temperature ($T=25\ °C$). Fig. 18 shows the results obtained with both conventional fixed step size P&O and proposed variable step size PO-GA algorithms.

The circled areas in Fig. 18 show the good tracking using the proposed PO-GA algorithm compared to P&O tracking in case of fast changing conditions. We can see that recovered energy in this case is very important using the proposed PO-GA algorithm compared to conventional P&O algorithm.

In the second test, we use a random temperature with fixed insolation ($S=1000\ \mathrm{W/m^2}$). Fig. 19 shows the results obtained with both conventional fixed step size P&O and proposed variable step size PO-GA algorithms.

It can be seen from Fig. 19 that our algorithm has a good tracking in case of temperature variation. Our algorithm outperforms the conventional fixed step size P&O algorithm in all variations cases. The accuracy of our algorithm is clearly confirmed.

The corresponding duty cycle generated for both algorithms in the tests (Figs. 18 and 19) is shown in Fig. 20.

From these results, the major contributions of this work are: the reduction of the ripple, the overshoot and the response time as well as the good ability of the proposed PO-GA algorithm to follow the MPP point especially in fast changing environment conditions, which resulting in an overall reduction of lost energy. The Table 7 summarizes the main contributions and improvements using the PO-GA proposed algorithm.

7. Conclusions

In this work, a new P&O variable step size based on genetic algorithm has been proposed for improving the maximum power point tracking in PV systems. The GA is used to tune the PID controller gains in order to optimize the variable step size needed by the P&O MPPT generating the PWM duty cycle to drive DC/DC boost converter. The new algorithm addresses the challenges associated with rapidly changing insolation levels. The simulation results have been presented validating the performance and functionality of the proposed algorithm. The simulations have been done using Simulink where the different aspects of the model design and parameters have been detailed. The simulation results were divided into transient, steady state, and dynamic response.

The first simulation environment was under variable step insolation levels (1000, 800, 600 and 400 $\mathrm{W/m^2}$). The output results of the proposed algorithm were compared to the output results of the P&O algorithm with fixed step-size. The comparison results were presented along with a table summarizing the efficiency improvement in both transient and steady-state responses. The proposed algorithm outperforms the conventional one considering all performance metrics (Ripple: 0.05% instead of 0.67%; overshoot: 1.3% instead of 9.69% and response time: 60 ms instead of 90 ms).

The second one, a random irradiation schemes were used to test the dynamic response of the algorithm, especially under rapidly changing atmospheric conditions. The output results were compared to the response of the fixed step-size P&O algorithm. The comparison figures have been presented and discussed showing the improvement in the dynamic response. The efficiency of the proposed algorithm is demonstrated in this case especially in energy recovered (1.4% every changing level). Algorithm robustness was verified using different schemes for temperature and insolation proving its ability to track the maximum power point in case of random and fast changing atmospheric conditions.

From these results, the major contribution of this work are: the reduction of the ripple, the overshoot and the response time as well as the good ability of the proposed PO-GA algorithm to follows the MPP point especially in fast changing environment conditions, which resulting in an overall reduction of lost energy.

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